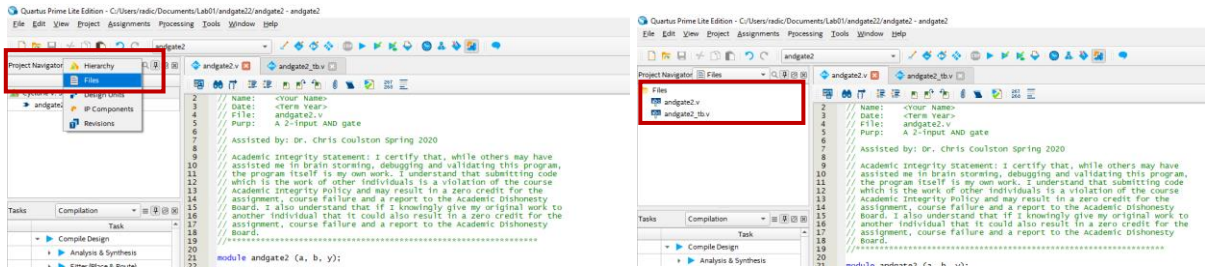
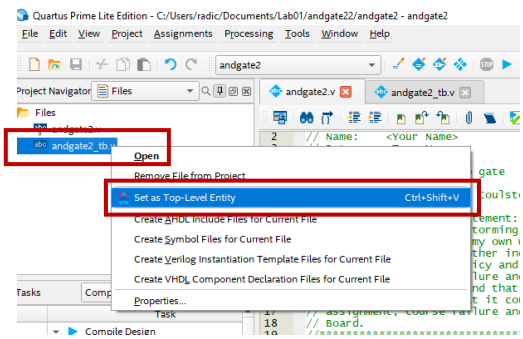
If you are planning on performing a simulation of your design, then the top-level entity should be a testbench. Inside the testbench should be an instantiation of your design as the unit under test(uut). Follow the snapshots and the red squares to work through a simulation. We will use the example of a two input and gate: andate2 which is also part1 of Lab1 assignment.

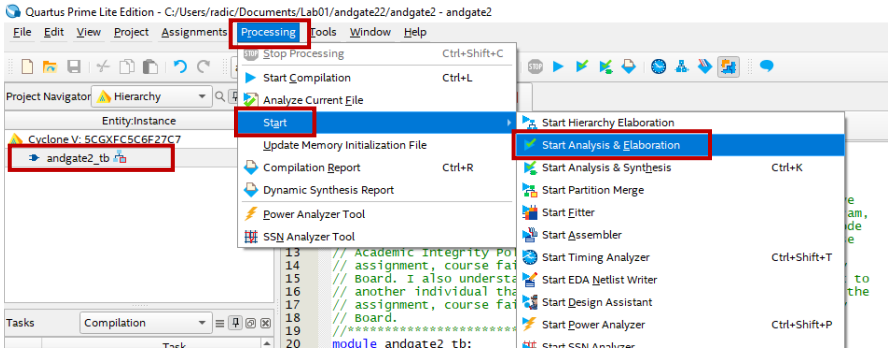
1. Click on the Files tab in the *Project Navigator* pane. You should see the two '.v' files we added to the project instead of andgate2 in the pane.



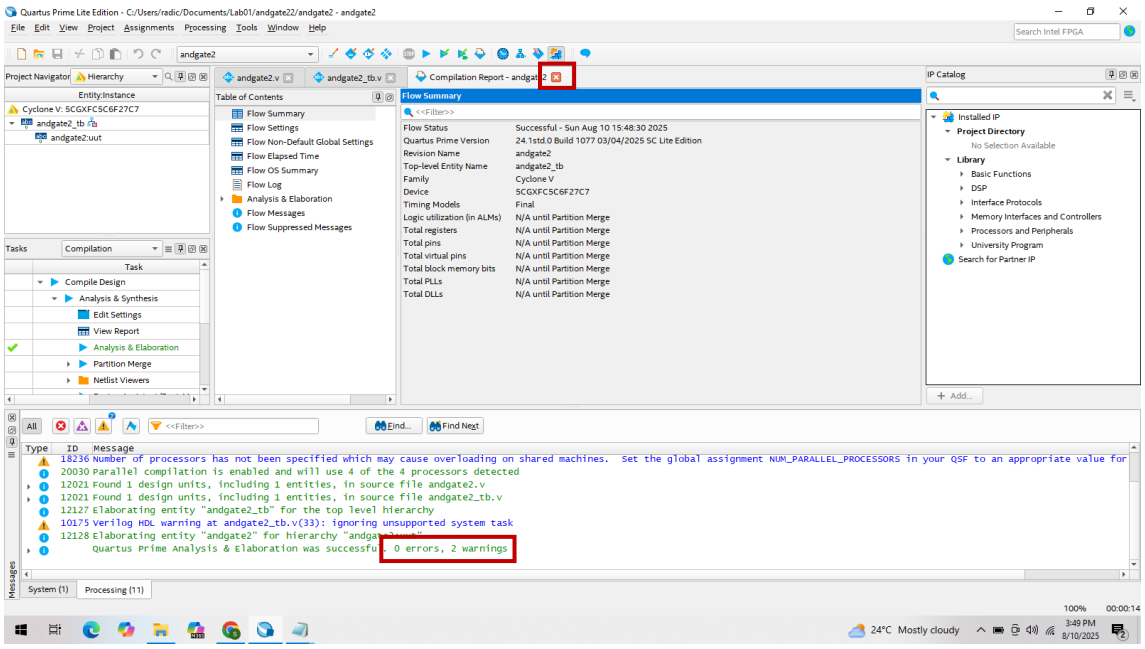
2. We want to make the testbench as the top-level entity. To do this: in the project navigator pane, right click on the test bench Verilog file `andgate2_tb.v` and select Set as Top-Level entity.



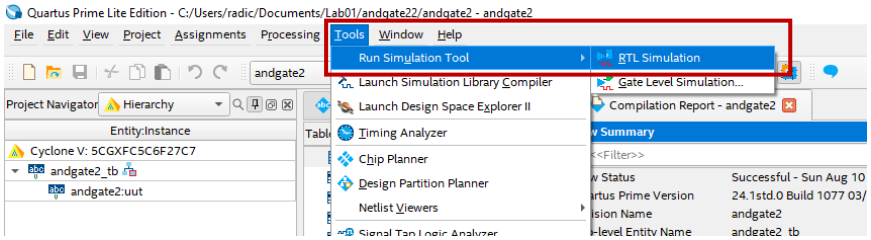
3. Click on the Hierarchy tab in the *Project Navigator* pane. You should now see andgate2_tb in the pane. In the main Quartus window, click on *Processing -> Start -> Start Analysis & Elaboration*. This may take some time, so be patient.



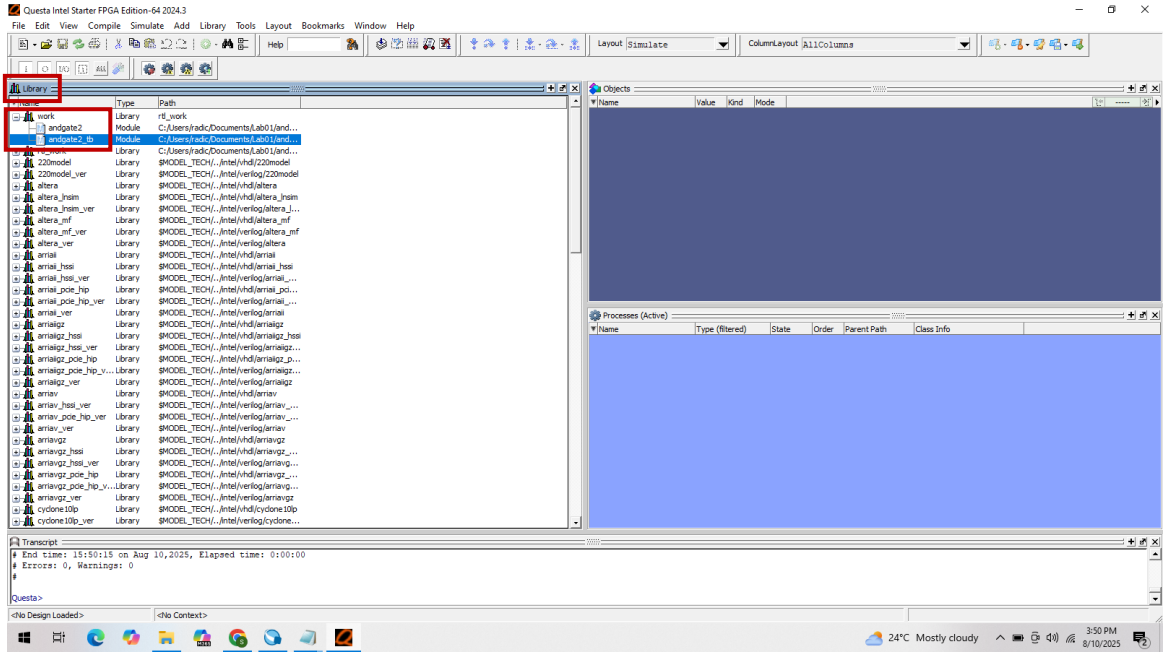
4. Once the compilation is complete, to move to the next step, there should be NO errors. It is also advisable to look at the warnings. Once everything looks good, you can close the compilation report by clicking on the x in the red box.



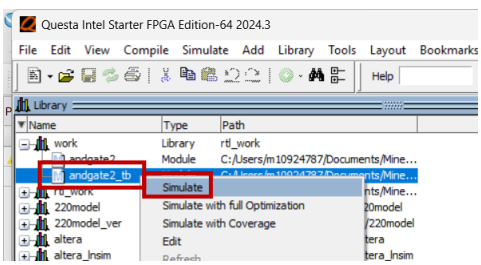
5. In the main Quartus window, click *Tools -> Run Simulation Tool -> RTL Simulation*. The Questa Intel FPGA program will launch. This may take a few moments, be patient.

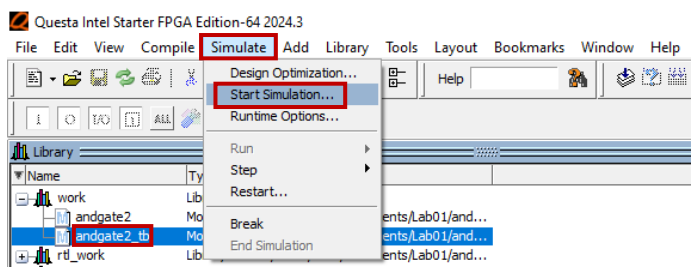


6. In Questa, you should see the following view when it first launches. The library pane should be the active pane. However, if not, find the *Library* pane in the menu option '*View*'. Expand the *work* library by clicking on the "+" at left.

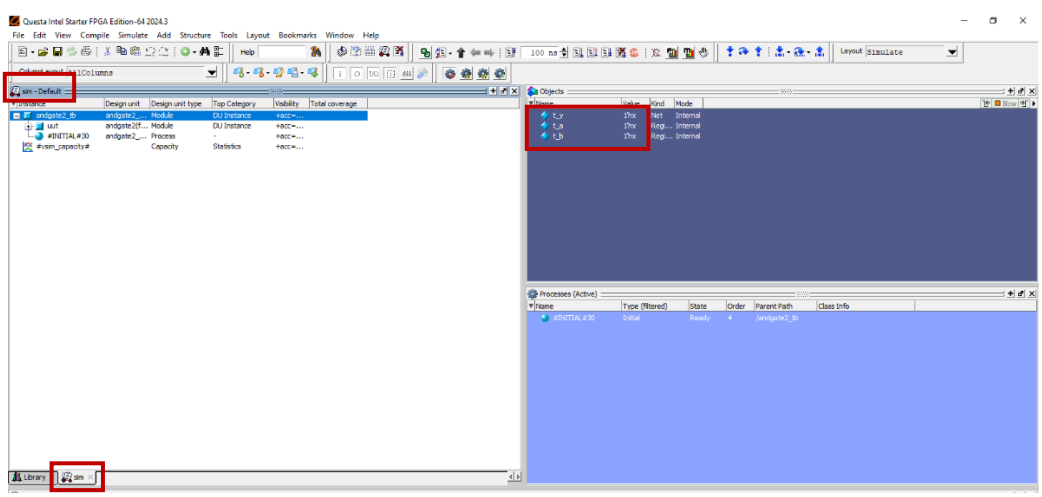


7. Right click on the top-level entity which is the testbench file `andgate2_tb.v` and click *Simulate*. You can also select the testbench file and click on *Simulate -> Start Simulation*.

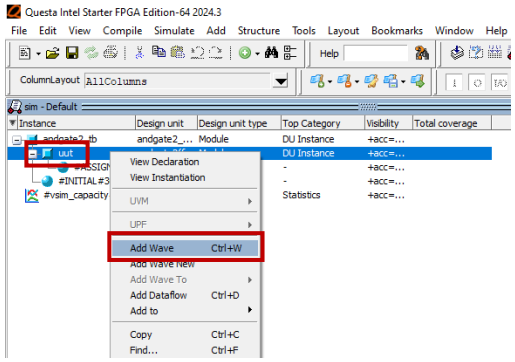




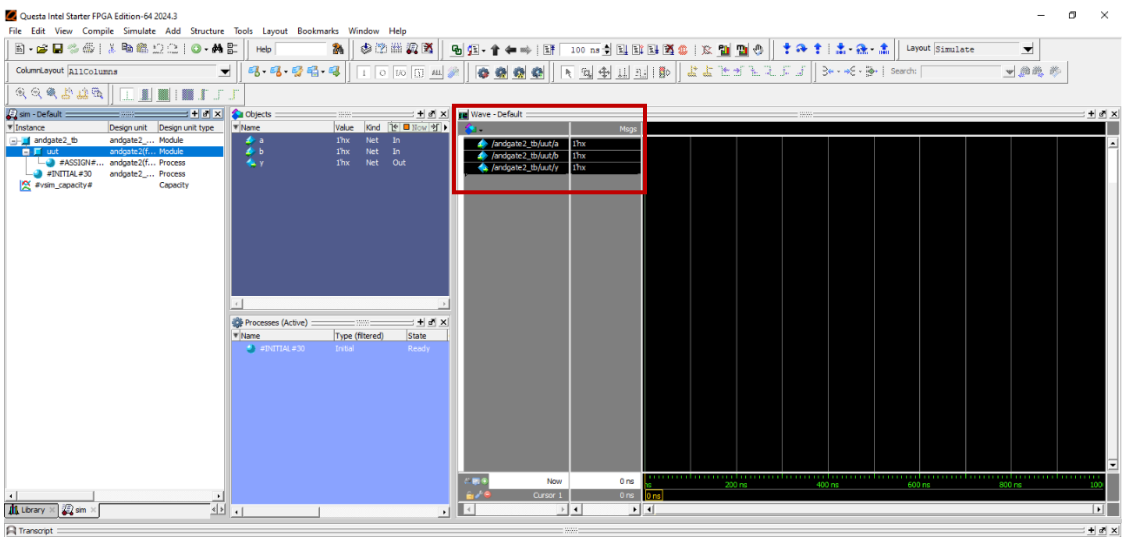
You should see the following view with a '*sim*' pane. Note how the testbench i/p and o/p signals have a value of '*x*':



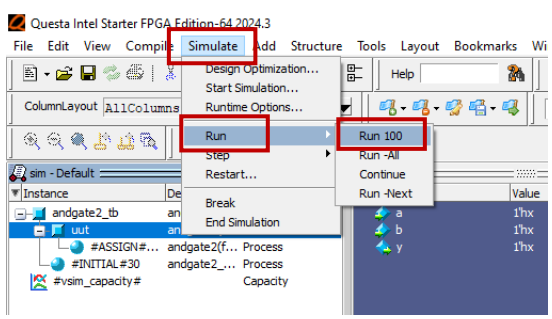
8. In the sim pane, right mouse click on uut and select *Add Wave*.



You should see the following view with the wave pane and the testbench signals added to the wave:

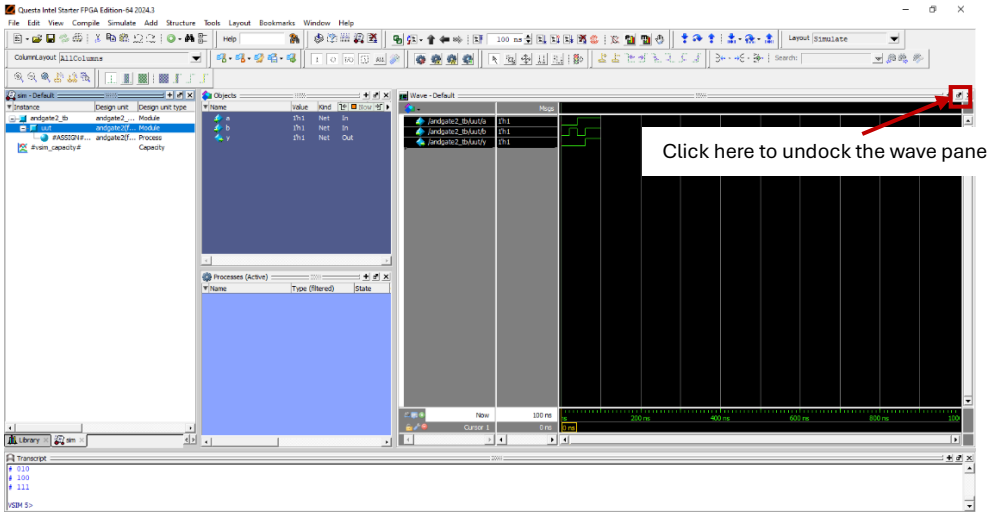


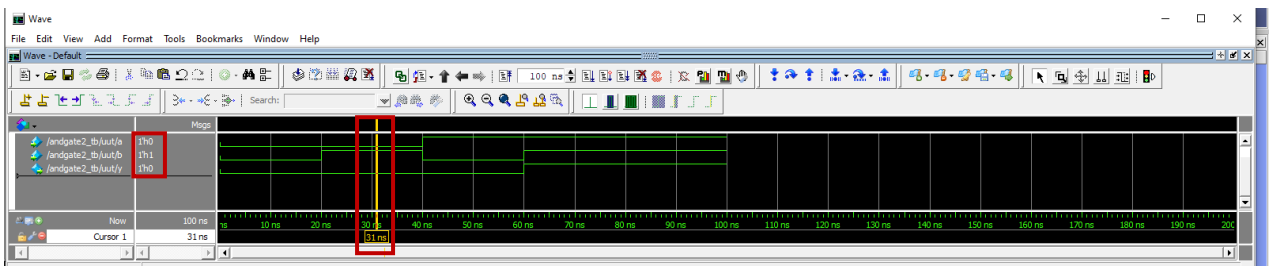
9. To generate the waveshapes for the signals, choose *Simulate -> Run -> Run 100*.



You should see the following view. Note that you can now see values associated with each signal and it is changing with time. This should correspond to how we have coded the testbench. It is a good idea to do a sanity check of your design here. We are designing an 'and gate' so if the i/p signals (*a,b)* are (1,0), the o/p *y* should be 0, etc. You can place the yellow cursor at any time by clicking in the wave pane to read the values

of all the signals at that time. Here, I have placed the cursor at 31 ns. At this time, a=0, b=1, and y =0.



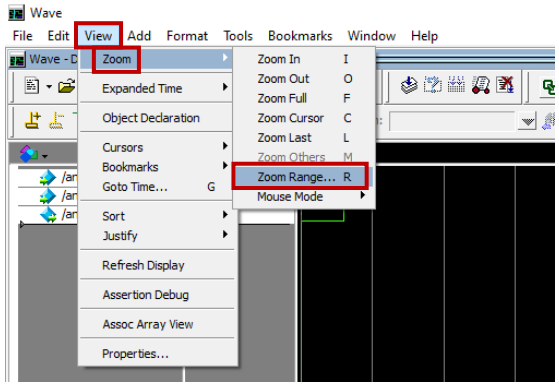


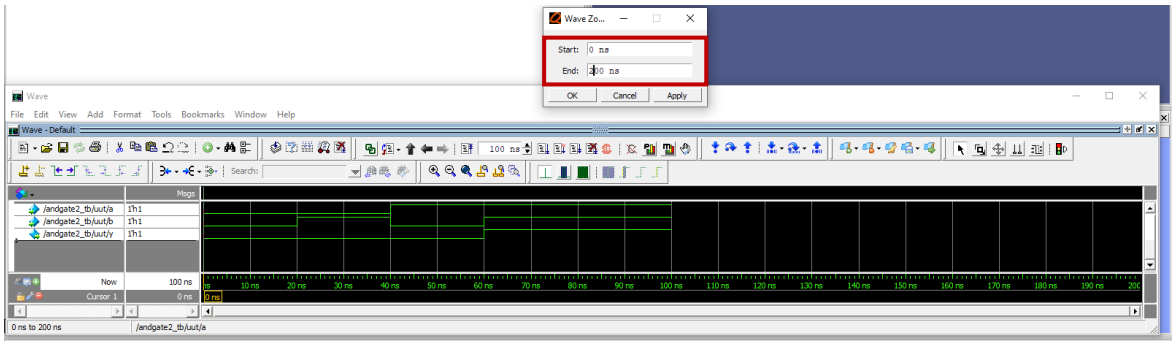
10. If you are asked to save the waveform. Perform the following steps:

    a. Undock the Wave pane. Resize the undocked Wave window vertically by grabbing its top edge and dragging down. Make the window tall enough to fit all the waves with a little room to spare.

    b. Re-order the waves so that the inputs are highest and outputs are lowest. Do this by grabbing their name and dragging it to the correct location.

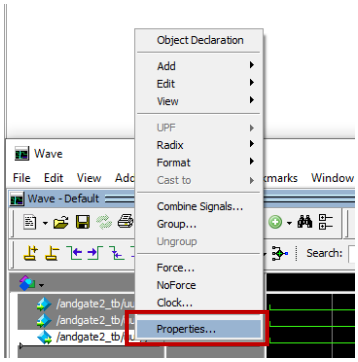
    c. To expand the waveshapes, click on *View -> Zoom –> Zoom Range*. Enter the range as 0-200 ns (note that we ran the simulation for 100 ns using Run100)

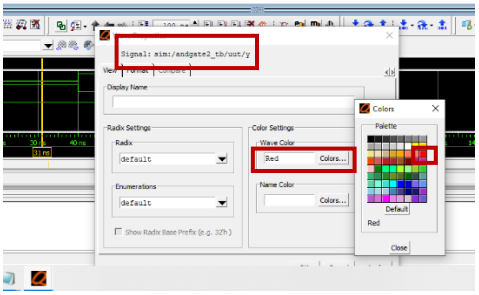




You can try other Zoom options for the wave to see which one allows best and clear view of the signals.

d. Color the output signals red. Right click on the signal that you want to change the color of and click on *Properties*. In this case we will change the o/p signal *y* to red. Leave the input signals, *a* and *b*, green. In the View tab of the Wave Properties pop-up, click the 'Colors...' button for Wave Color and choose Red, click Close, then OK.





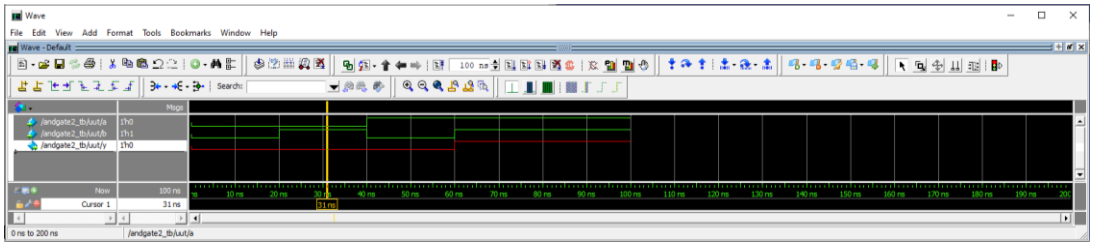
In later labs we will follow this color convention:

i/p signals – Green

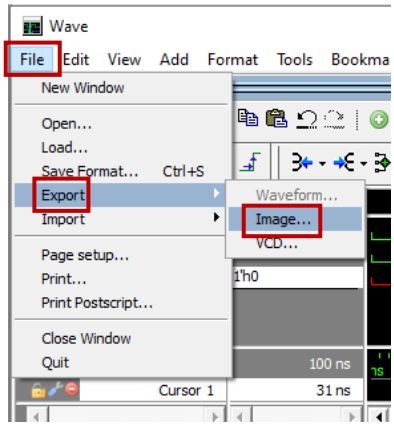
o/p signals – Red

intermediate signals – Yellow

You should see the following view:

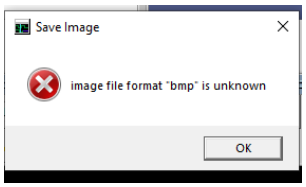


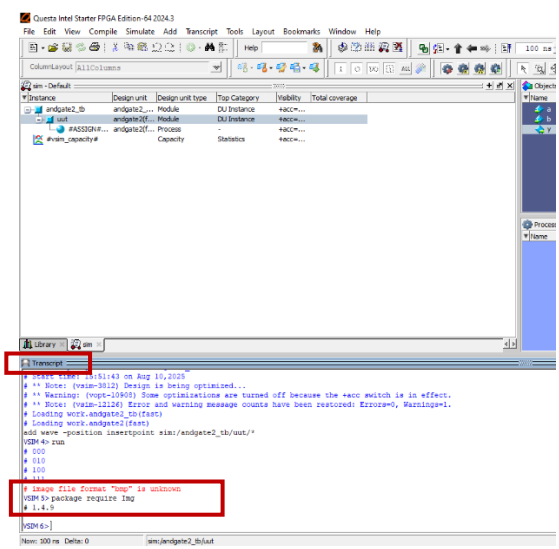
e. To save the waveshape, you can use any snipping tool or print screen. We can also export the waveshape as an image file. For this, click File -> Export -> Image.



f. Navigate to your project directory, provide a File name, then click Save as a bitmap or png.

g. If you see an error pop up, it could be because of a missing package in Questa. To fix this problem, in the Transcript pane, type: `package require Img`. Press enter and try steps e and f again.





11. Close Questa. Do not save wave commands.